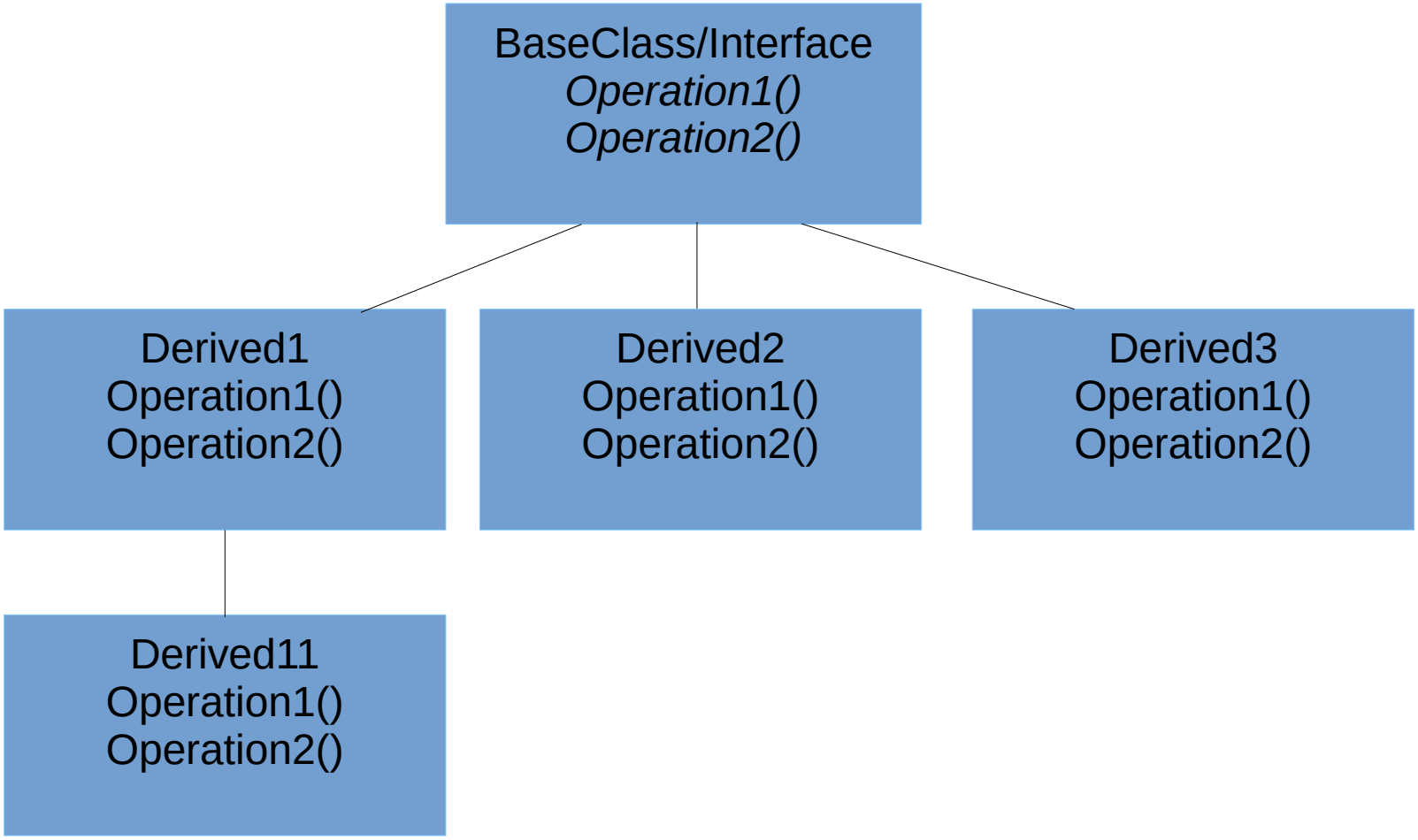
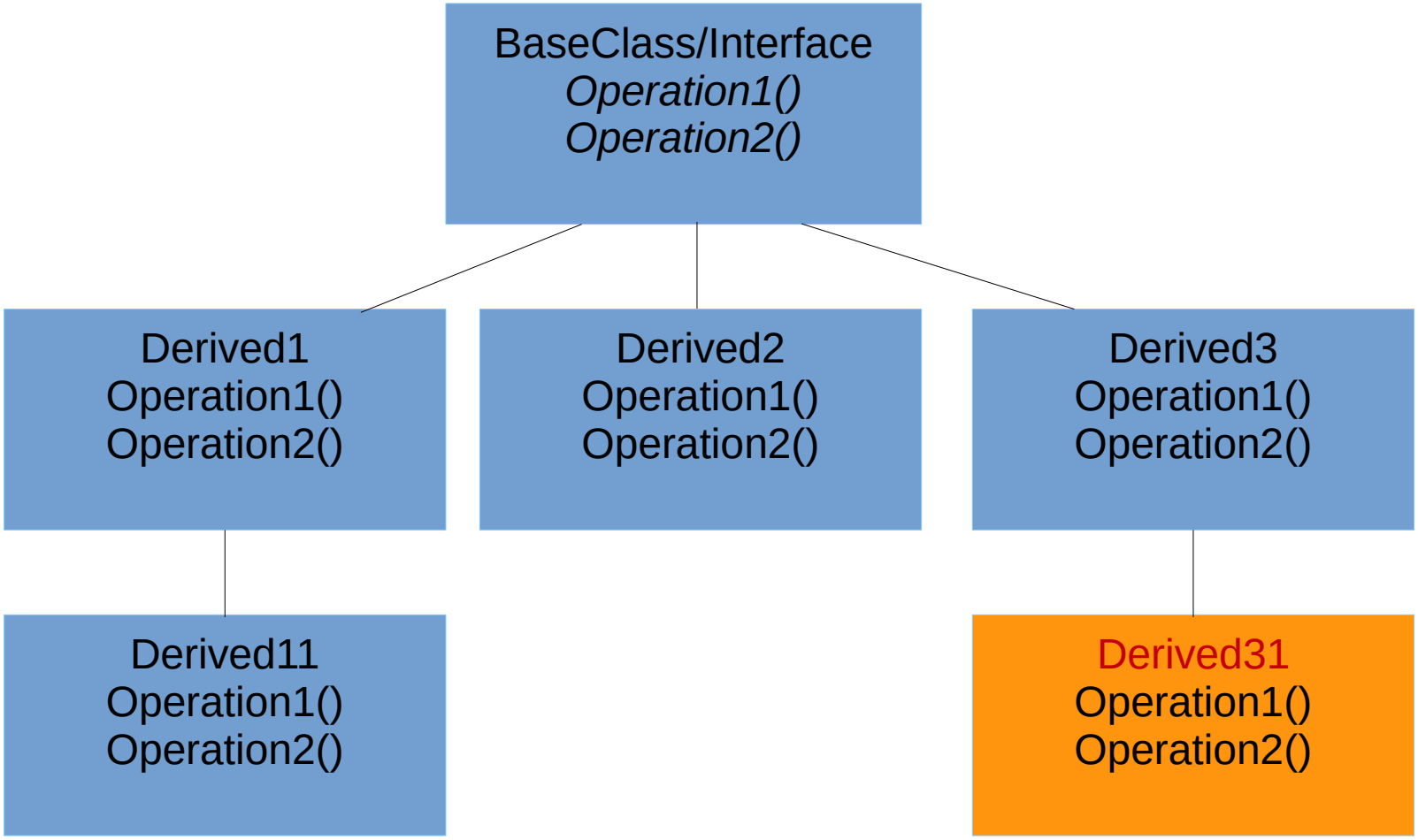
STL

- Expression problem
- Generic programming
- An example – inserters, iterator-adapters, functors
- Efficiency
- Memory consumption characteristics
- Array and forward_list in C++11
- Unordered containers in C++11

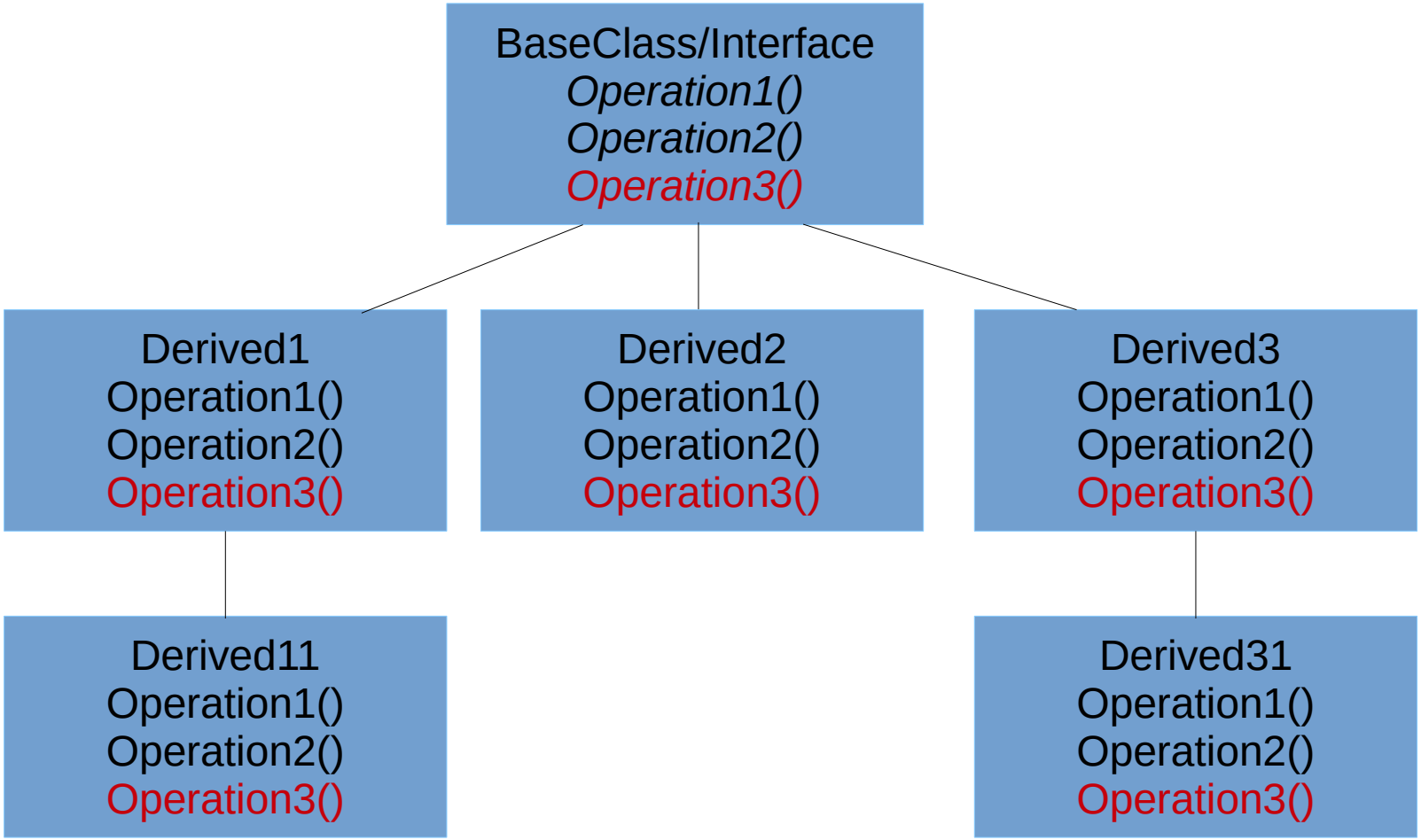
The expression problem



The expression problem



The expression problem



The expression problem

- Philip Wadler: expression problem mail, 1990
- Shriram Krishnamurthi, Matthias Felleisen, Daniel P. Friedman: "Synthesizing Object-Oriented and Functional Design to Promote Re-Use", 1998

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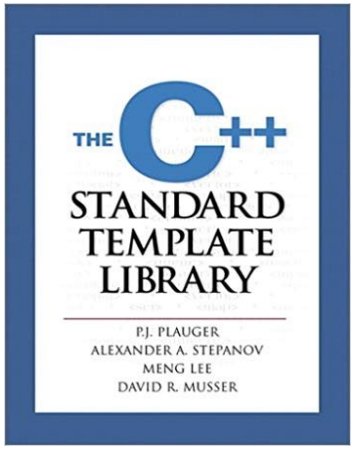
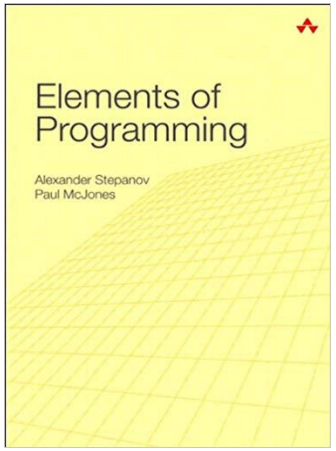
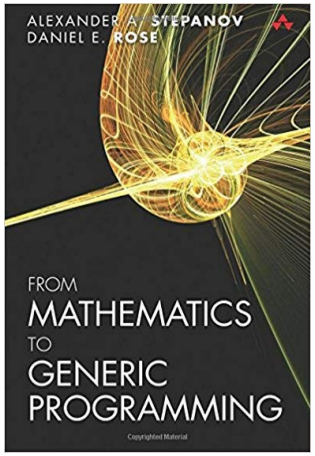
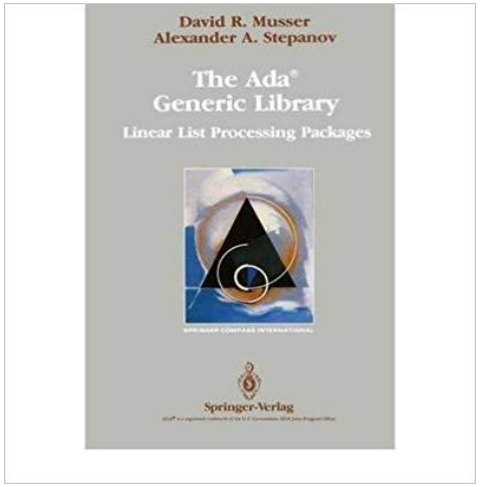
- Aleksey Stepanov: generic programmig, 1985









The STL solution

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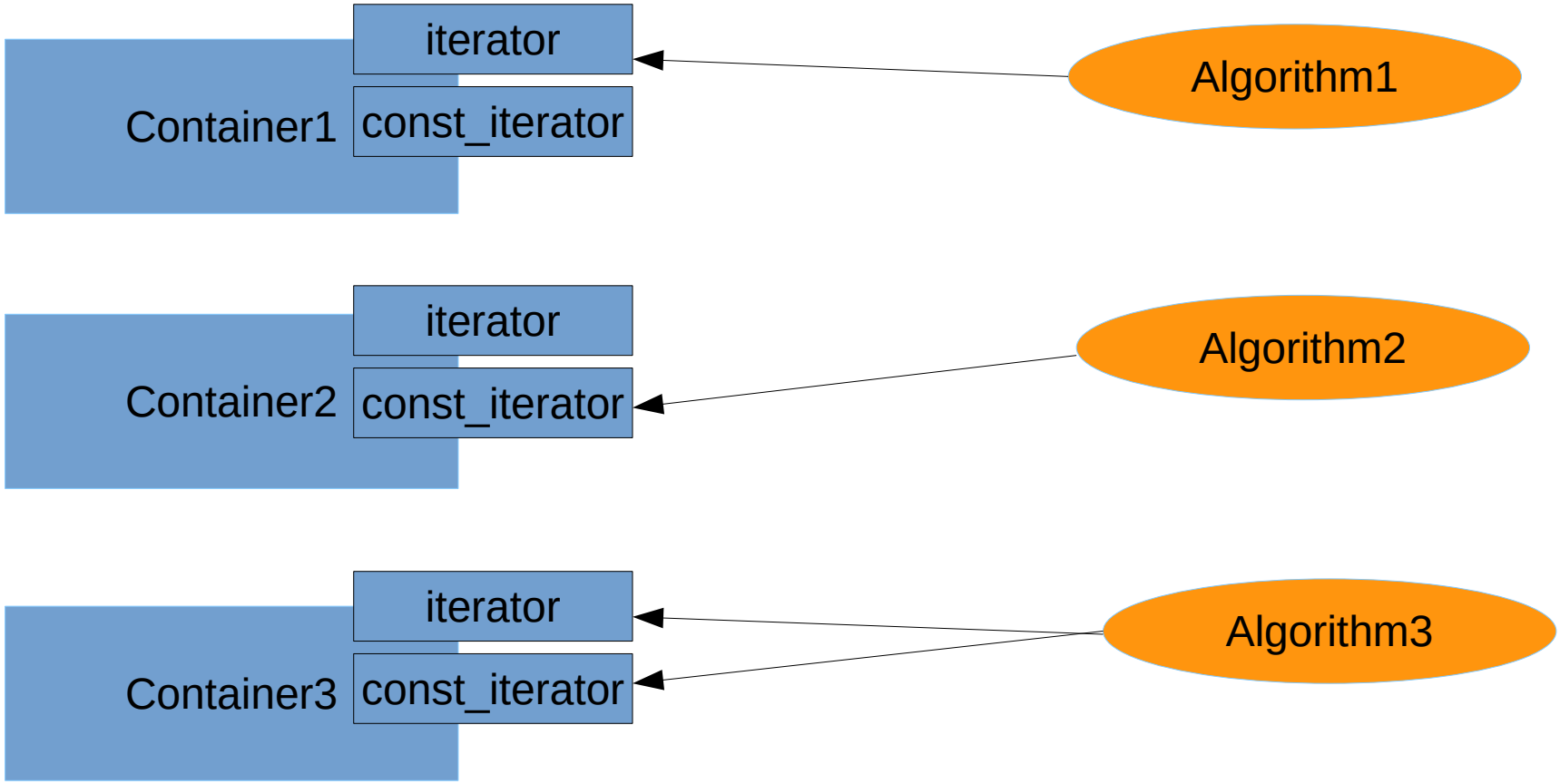
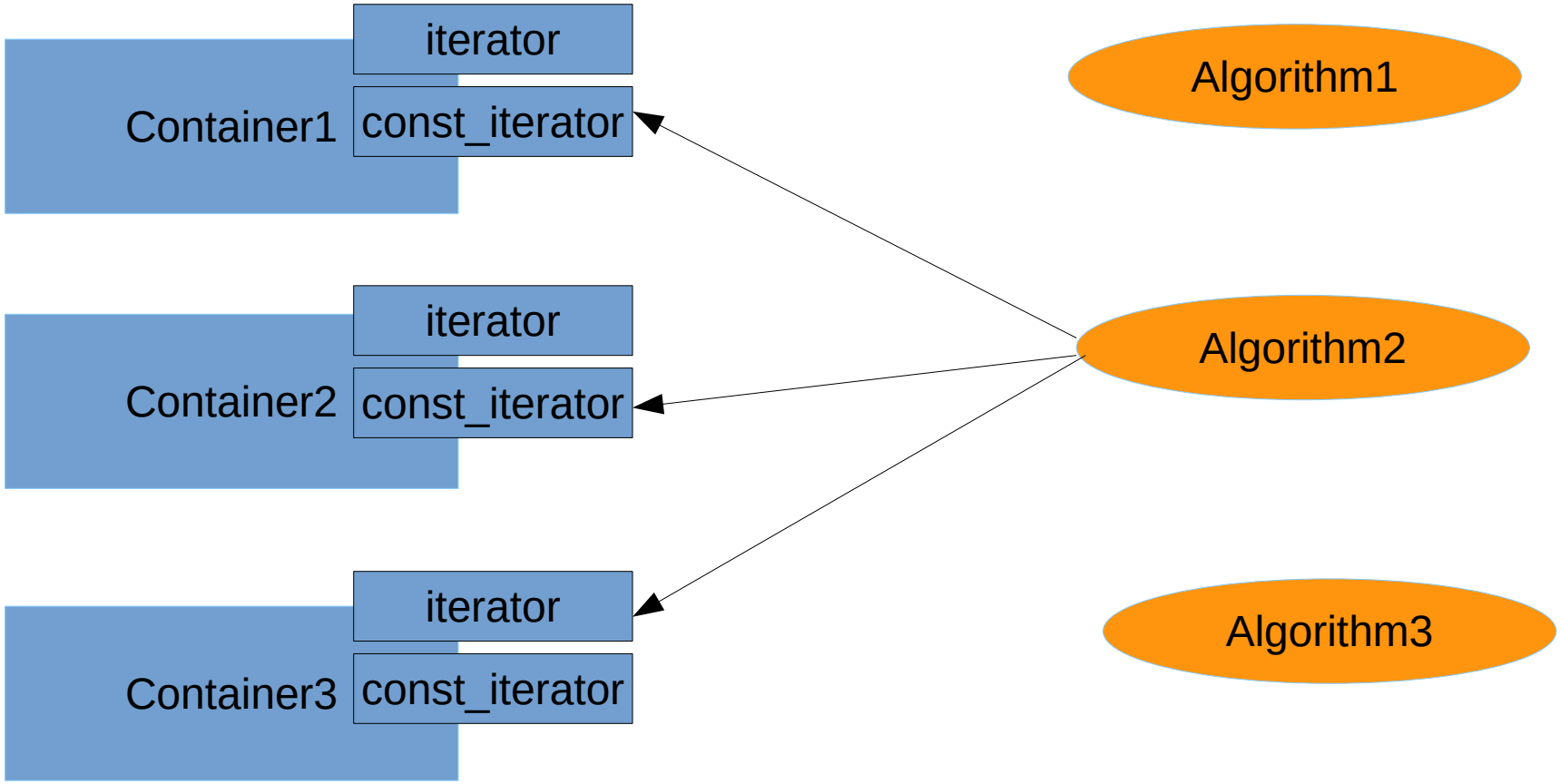
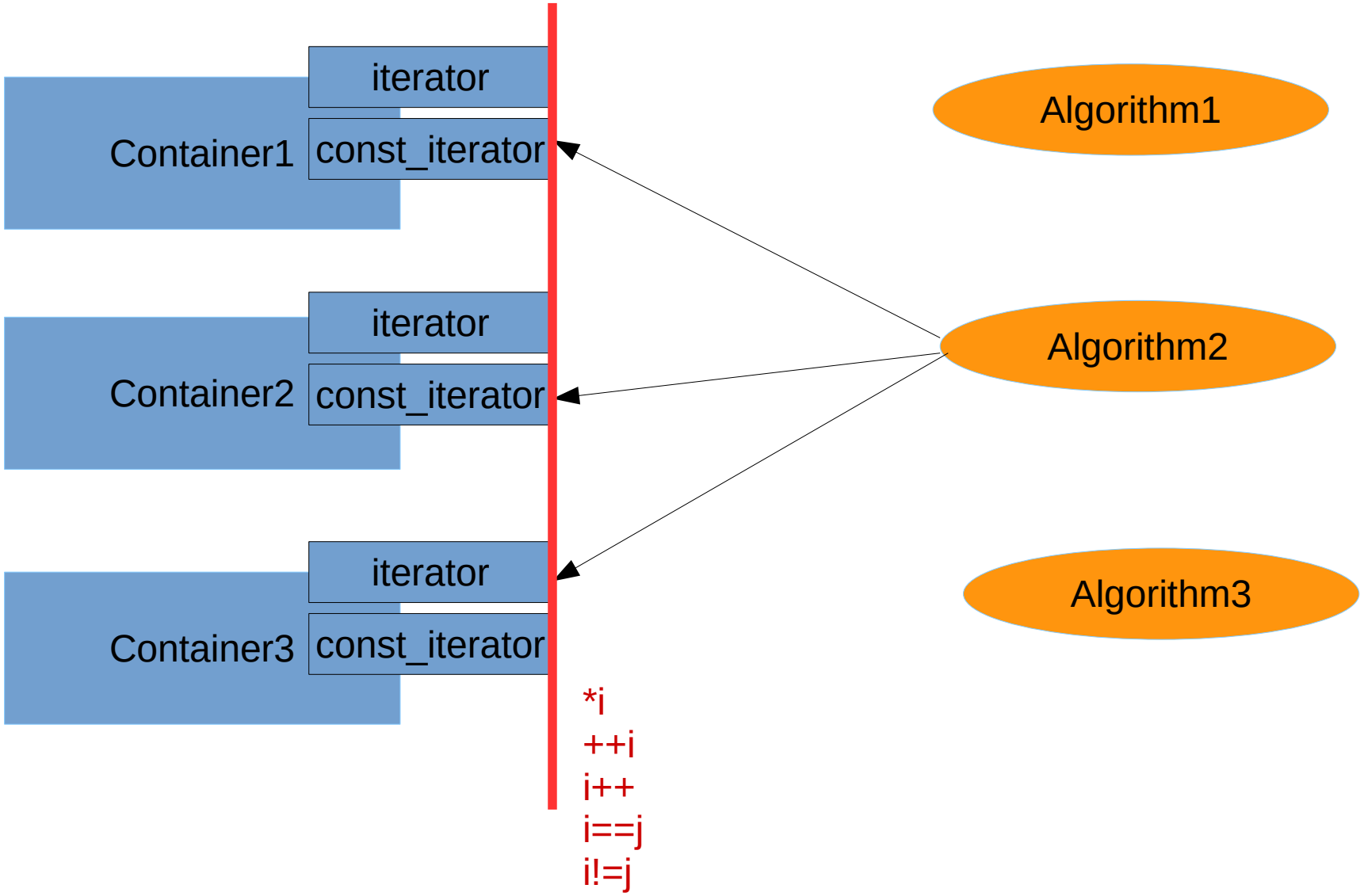


The STL solution



The STL solution



How to implement

```
int t[] = { 1, 3, 5, ... };

// find the first occurrence of value 55
int *pi = find( t, t+sizeof(t)/sizeof(t[0]), 55);

if ( pi )
{
  *pi = 56
}

// a very specific algorithm: works only on integer arrays
int *find( int *begin, int *end, int x)
{
  while ( begin != end )
  {
    if ( *begin == x )
    {
      return begin;
    }
    ++begin;
  }
  return nullptr;
}
```

Simple solution

```
int t[] = { 1, 3, 5, ... };

// find the first occurrence of value 55
int *pi = find( t, t+sizeof(t)/sizeof(t[0]), 55);

if ( pi )
{
  *pi = 56
}
```

Template based

```
double t[] = { 1.0, 3.14, 5.55, ... };

// find the first occurrence of a value
double *pi = find( t, t+sizeof(t)/sizeof(t[0]), 55.5);

if ( pi )
{
  *pi = 56.5
}

// Templated algorithm
template <typename T>
T *find( T *begin, T *end, const T& x)
{
  while ( begin != end )
  {
    if ( *begin == x )
    {
      return begin;
    }
    ++begin;
  }
  return nullptr;
}
```

Iterator based

```
std::list<int> li = { 1, 3, 5, ... };

// find the first occurrence of value 55
auto it = find( li.begin(), li.end(), 55);

if ( li.end() != it )
{
  *it = 56
}

// Iterator based algorithm
template <typename It, typename T>
It find( It begin, It end, const T& x)
{
  while ( begin != end )
  {
    if ( *begin == x )
    {
      return begin;
    }
    ++begin;
  }
  return end;  // not nullptr
}
```

Universal usage

```
std::list<int> li = { 1, 3, 5, ... };
std::vector<double> vd = { 1.0, 3.3, 5.5, ... };

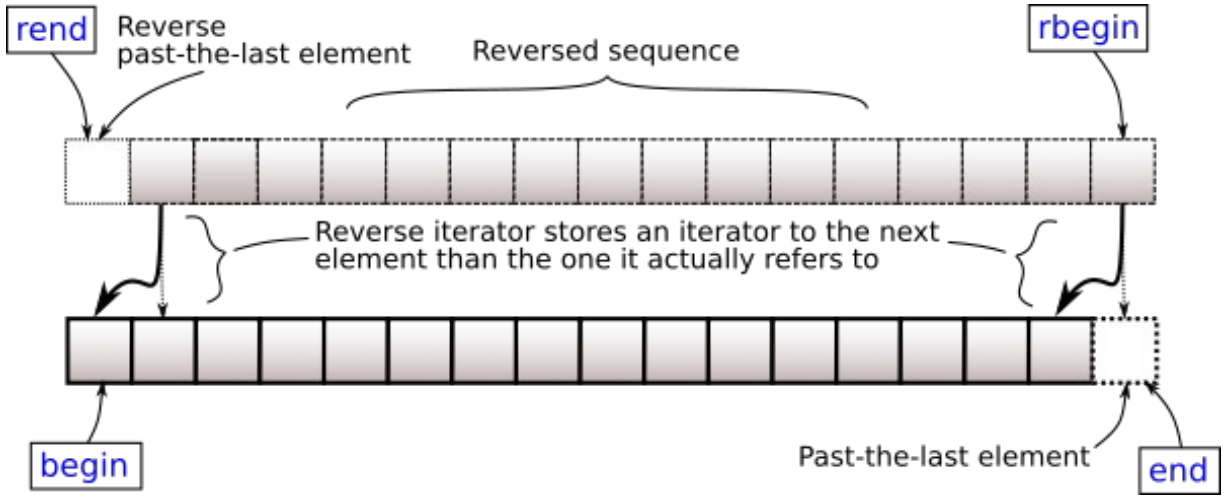
template <typename Container>
auto generic_find( const Container& c, typename Container::value_type& v)
{
  return std::find( c.begin(), c.end(), v);
}
```

- ```
  template<class C>
  typename C::value_type generic_sum(const C& c)
  {
    typename C::value_type s{};                  // value construct
    typename C::const_iterator it = c.begin();  // auto it = c.begin()

    while ( it != c.end() )
    {
      s += *it;        // requires operator+= on C::value_type
      ++it;
    }
    return s;
  }

  auto i = generic_find(li, 5);
  auto d = generic_sum(vd);
  ```
  ```

Reverse iterator




```
template<class C>
typename C::iterator find_last(C& c, const typename C::value_type& v)
{
  typename C::reverse_iterator p = c.rbegin(); // view sequence in revers
  while ( p != c.rend() )
  {
    if ( *p == v )
    {
      typename C::iterator i = p.base();
      return --i;
    }
    ++p;            // note: increment, not decrement (--)
  }
  return c.end();   // use c.end() to indicate "not found”
}
```

Iterators are const safe

```
//  C++11
std::vector<int>  v1 = { 1, 2, 3, 4, ... };
auto i = std::find( v1.begin(), v1.end(), 3);
// i is vector::iterator

const std::vector<int>  v2 = { 1, 2, 3, 4, ... };
auto j = std::find( v2.begin(), v2.end(), 3);
// j is vector::const_iterator

auto k = std::find( v1.cbegin(), v1.cend(), 3);
// k is vector::const_iterator


auto j2 = std::find( std::begin(v1), std::end(v1), 3);
// j2 is vector::iterator

auto k2 = std::find( std::cbegin(v1), std::cend(v1), 3);
// k2 is vector::const_iterator
```

Functors

```
std::list<int> li = { 1, 3, 5, ... };

// find the third occurrence of value less than 55
auto it = find_if( li.begin(), li.end(), ?);

if ( li.end() != it )
{
  *it = 56
}

// Iterator based algorithm
template <typename It, typename Pred>
It find_if( It begin, It end, Pred p)
{
  while ( begin != end )
  {
    if ( p(*begin) )
    {
      return begin;
    }
    ++begin;
  }
  return end;
}
```

Functors

```
std::list<int> li = { 1, 3, 5, ... };

// find the third occurrence of value less than 55
bool less55_3rd(int x)
{
  static int cnt = 0;
  if ( x < 55 )    ++cnt;
  return 3 == cnt;
}

// Iterator based algorithm
template <typename It, typename Pred>
It find_if( It begin, It end, Pred p)
{
  while ( begin != end )
  {
    if ( p(*begin) )  // calls less55_3rd(*begin)
    {
      return begin;
    }
    ++begin;
  }
  return end;
}
```

Functors

```
std::list<int> li = { 1, 3, 5, ... };


bool less55_3rd(int x)
{
  static int cnt = 0;
  if ( x < 55 )    ++cnt;
  return 3 == cnt;
}

// find the third occurrence of value less than 55
auto it = find_if( li.begin(), li.end(), less55_3rd);

if ( li.end() != it )
{
  *it = 56;

}
```

Functors

```
std::list<int> li = { 1, 3, 5, ... };


bool less55_3rd(int x)
{
  static int cnt = 0;
  if ( x < 55 )    ++cnt;
  return 3 == cnt;
}

// find the third occurrence of value less than 55
auto it = find_if( li.begin(), li.end(), less55_3rd);

if ( li.end() != it )
{
  *it = 56;
   it = find_if( ++it, li.end(), less55_3rd);  // works?
}
```

Functors

```
struct less55_3rd
{
  less55_3rd() : cnt(0) { }
  bool operator()(int x)
  {
    if ( x < 55 )    ++cnt;
    return 3 == cnt;
  }
private:
  int cnt;
};

template <typename It, typename Pred>
It find_if( It begin, It end, Pred p)
{
  while ( begin != end )
  {
    if ( p(*begin) )  // calls p.operator()(*begin)
    {
      return begin;
    }
    ++begin;
  }
  return end;
}
```

Functors

```
struct less55_3rd
{
  less55_3rd() : cnt(0) { }
  bool operator()(int x)
  {
    if ( x < 55 )    ++cnt;
    return 3 == cnt;
  }
private:
  int cnt;
};

// find the third occurrence of value less than 55
auto it = find_if( li.begin(), li.end(), less55_3rd{});


if ( li.end() != it )
{
  *it = 56;
   it = find_if( ++it, li.end(), less55_3rd{});  // new object, cnt = 0
}
```

Functors

```
struct less55_3rd
{
  less55_3rd() : cnt(0) { }
  bool operator()(int x)
  {
    if ( x < 55 )    ++cnt;
    return 3 == cnt;
  }
private:
  int cnt;
};

// find the third occurrence of value less than 55
auto it = find_if( li.begin(), li.end(), [](int x) { static int cnt = 0;
                                                     if(x < 55) ++cnt;
                                                     return 3 == cnt;} );
if ( li.end() != it )
{
  *it = 56;
   it = find_if( ++it, li.end(), [](int x) { ... }); // new lambda, cnt = 0
}
```

Functors

```
struct less55_3rd
{
  less55_3rd() : cnt(0) { }
  bool operator()(int x)
  {
    if ( x < 55 )    ++cnt;
    return 3 == cnt;
  }
private:
  int cnt;
};

// find the third occurrence of value less than 55
auto it = find_if( li.begin(), li.end(), [cnt=0](int x) { // since C++14
                                                          if(x < 55) ++cnt;
                                                          return 3 == cnt;} );
if ( li.end() != it )
{
  *it = 56;
   it = find_if( ++it, li.end(), [cnt=0](int x) { ... }); // new lambda
}
```

Functors

```
template <typename T>
struct less_Nth
{
  less_Nth(const T& t, int n) : value(t), nth(n), cnt(0) { }
  bool operator()(const T& x)
  {
    if ( x < value )    ++cnt;
    return nth == cnt;
  }
private:
  T   value;
  int nth;
  int cnt;
};

// find the fifth occurrence less than value 3.14
auto it = find_if( li.begin(), li.end(), less_Nth<double>{3.14,5});

if ( li.end() != it )
{
  *it = 2.178;
   it = find_if( ++it, li.end(), less_Nth<double>{99.9,7});
}
```

Example: merge two files

```
#include <iostream>
#include <fstream>
#include <string>

using namespace std;

int main()   // simple merge:
{
    string s1, s2;
    ifstream f1("file1.txt");
    ifstream f2("file2.txt");

    f1 >> s1; f2 >> s2;
    while (f1 || f2)
    {
        if (f1 && ((s1 <= s2) || !f2))
        {
            cout << s1 << endl;
            f1 >> s1;
        }
        if (f2 && ((s1 >= s2) || !f1))
        {
            cout << s2 << endl;
            f2 >> s2;
        }
    }
    return 0;
}
```

http://www.caesar.elte.hu/progmat/kultura/humor/zenfothi.html

https://terebess.hu/zen/mesterek/fothi-akos.html

Example: naïve STL usage

```
#include <iostream>
#include <fstream>
#include <string>
#include <algorithm>    // merge( b1, e1, b2, e2, b3 [,opc_rend])
#include <vector>

using namespace std;
int main()
{
    ifstream if1("file1.txt");
    ifstream if2("file2.txt");

    string s;
    vector<string> v1;
    while ( if1 >> s ) v1.push_back(s); // do we have enough memory?
    vector<string> v2;
    while ( if2 >> s ) v2.push_back(s); // do we have enough memory?

    // allocate the elements for the result, do we have enough memory?
    vector<string> v3(v1.size() + v2.size());   // constructs empty strings...

    // *b3++ = *b1 < *b2 ? *b1++ : *b2++
    merge( v1.begin(),v1.end(),v2.begin(),v2.end(),v3.begin());

    for ( int i = 0; i < v3.size(); ++i)
        cout << v3[i] << endl;

    return 0;
}
```

std::merge

```
template<class InputIt1, class InputIt2, class OutputIt>
OutputIt merge(InputIt1 first1, InputIt1 last1,
               InputIt2 first2, InputIt2 last2,
               OutputIt d_first)
{
    for (; first1 != last1; ++d_first) {
        if (first2 == last2) {
            return std::copy(first1, last1, d_first);
        }
        if (*first2 < *first1) {
            *d_first = *first2;
            ++first2;
        } else {
            *d_first = *first1;
            ++first1;
        }
    }
    return std::copy(first2, last2, d_first);
}
```

std::merge

```
template<class InputIt1, class InputIt2, class OutputIt>
OutputIt merge(InputIt1 first1, InputIt1 last1,
               InputIt2 first2, InputIt2 last2,
               OutputIt d_first, Compare comp)
{
    for (; first1 != last1; ++d_first) {
        if (first2 == last2) {
            return std::copy(first1, last1, d_first);
        }
        If ( comp(*first2,*first1) ) {
            *d_first = *first2;
            ++first2;
        } else {
            *d_first = *first1;
            ++first1;
        }
    }
    return std::copy(first2, last2, d_first);
}
```

Example: inserters

```
#include <iostream>
#include <fstream>
#include <string>
#include <algorithm>
#include <vector>

using namespace std;

int main()
{
    ifstream if1("file1.txt");
    ifstream if2("file2.txt");

    string s;
    vector<string> v1;
    while ( if1 >> s ) v1.push_back(s);
    vector<string> v2;
    while ( if2 >> s ) v2.push_back(s);
    vector<string> v3;
    v3.reserve( v1.size() + v2.size() ); // alloc buffer but do not construct, size == 0

    merge(v1.begin(),v1.end(),v2.begin(),v2.end(),back_inserter(v3)); // v3.push_back(*c)

    for ( int i = 0; i < v3.size(); ++i)
        cout << v3[i] << endl;

    return 0;
}
```

Example: inserters

```
template<typename _Cont>
class back_insert_iterator : public
      iterator<output_iterator_tag,void,void,void,void>
{
  back_insert_iterator&
  operator=(const typename _Cont::value_type& __value)
  {   // same overloaded to reference and rvalue ref
    container->push_back(__value);
    return *this;
  }
  back_insert_iterator& operator*()   { return *this; }
  back_insert_iterator& operator++()  { return *this; }
  back_insert_iterator operator++(int){ return *this; }
protected:
  _Cont* container;
};
```

```
#include <iostream>
#include <fstream>
#include <string>
#include <algorithm>
#include <vector>

using namespace std;

int main()
{
    ifstream if1("file1.txt");
    ifstream if2("file2.txt");

    string s;
    vector<string> v1;
    while ( if1 >> s ) v1.push_back(s);
    vector<string> v2;
    while ( if2 >> s ) v2.push_back(s);
    vector<string> v3;
    v3.reserve( v1.size() + v2.size() ); // alloc buffer but do not construct, size == 0

    merge(v1.begin(),v1.end(),v2.begin(),v2.end(),back_inserter(v3)); // v3.push_back(*c)

    for ( int i = 0; i < v3.size(); ++i)
        cout << v3[i] << endl;

    return 0;
}
```

Example: stream iterators

```
#include <iostream>
#include <fstream>
#include <string>
#include <algorithm>
#include <iterator>     // input- and output-iterators

using namespace std;

int main()
{
    ifstream if1("file1.txt");
    ifstream if2("file2.txt");

    // istream_iterator(if1) -> if1 >> *current
    // istream_iterator() -> EOF
    // ostream_iterator(of,x) -> of << *current << x
    merge( istream_iterator<string>(if1), istream_iterator<string>(),
           istream_iterator<string>(if2), istream_iterator<string>(),
           ostream_iterator<string>(cout,"\n") );

    return 0;
}
```

Example: comparator

```
#include <iostream>
#include <fstream>
#include <string>
#include <cctype>
#include <algorithm>
#include <iterator>

struct my_less  // function object: "functor"
{
    bool operator()(const std::string& s1, const std::string& s2)  {
        std::string us1 = s1;
        std::string us2 = s2;
        transform( s1.begin(), s1.end(), us1.begin(), toupper);  // TODO: use <locale>
        transform( s2.begin(), s2.end(), us2.begin(), toupper);
        return us1 < us2;
    }
};

int main()
{
    ifstream if1("file1.txt");
    ifstream if2("file2.txt");

    merge( istream_iterator<string>(if1), istream_iterator<string>(),
           istream_iterator<string>(if2), istream_iterator<string>(),
           ostream_iterator<string>(cout,"\n"), my_less() );
    return 0;
}
```

Example: template comparator

```
template <typename T>
class distr
{
public:
    distr(int l, int r, bool fl = true) : left_(l), right_(r), from_left_(fl), cnt_(0) { }
    // formal reasons: "compare" has two parameters of type T
    bool operator()( const T&, const T&)   {
        bool ret = from_left_;   // from_left_ is “smaller” currently
        const int max = from_left_ ? left_ : right_;
        if ( ++cnt_ == max )
        {
            cnt_ = 0;
            from_left_ = ! from_left_;
        }
        return ret;
    }
private:
    const int left_;  // read left_ element from left
    const int right_; // read right_ element from right
    int from_left_;   // start from left
    int cnt_;
};

   // ...
   istream_iterator<string>(if2), istream_iterator<string>(),
   ostream_iterator<string>(cout,"\n"), distr<std::string>(left,right) );
   // ...
```

Vector vs Associative containers

```
#include <iostream>
#include <string>
#include <algorithm>
#include <set>
using namespace std;

int main()   /* print unique sorted elems  */
{
    set<string> coll( istream_iterator<string>(cin), istream_iterator<string>());
    copy( coll.begin(), coll.end(), ostream_iterator<string>(cout, "\n"));
}
```

```
#include <iostream>
#include <string>
#include <algorithm>
#include <vector>
using namespace std;

int main()   /* print unique sorted elems  */
{
    vector<string> coll( istream_iterator<string>(cin), istream_iterator<string>());
    sort (coll.begin(), coll.end());   // sort elements
    unique_copy (coll.begin(), coll.end(), ostream_iterator<string>(cout, "\n"));
}
```

Vector vs Associative containers

```
#include <iostream>
#include <string>
#include <algorithm>
#include <set>
using namespace std;

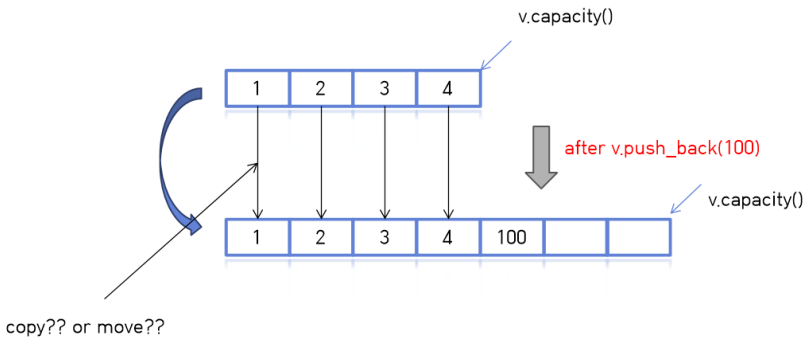
int main()   /* print unique sorted elems  */
{
    set<string> coll( istream_iterator<string>(cin), istream_iterator<string>());
    copy( coll.begin(), coll.end(), ostream_iterator<string>(cout, "\n"));
}
```

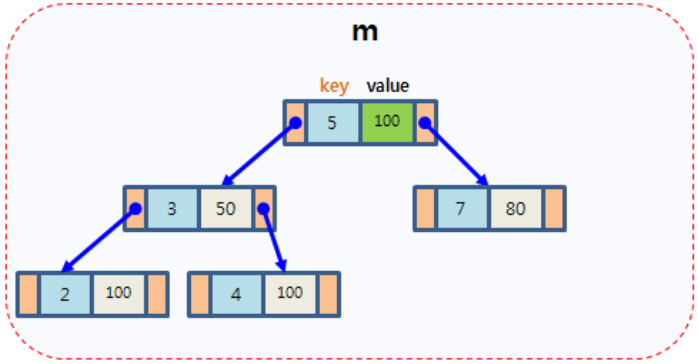
```
#include <iostream>
#include <string>
#include <algorithm>
#include <vector>
using namespace std;

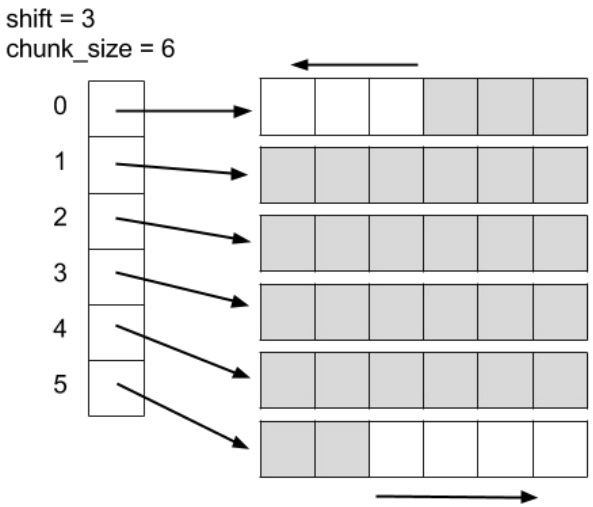
int main()   /* print unique sorted elems  */
{
    vector<string> coll( istream_iterator<string>(cin), istream_iterator<string>());
    sort (coll.begin(), coll.end());   // sort elements
    unique_copy (coll.begin(), coll.end(), ostream_iterator<string>(cout, "\n"));
}
```

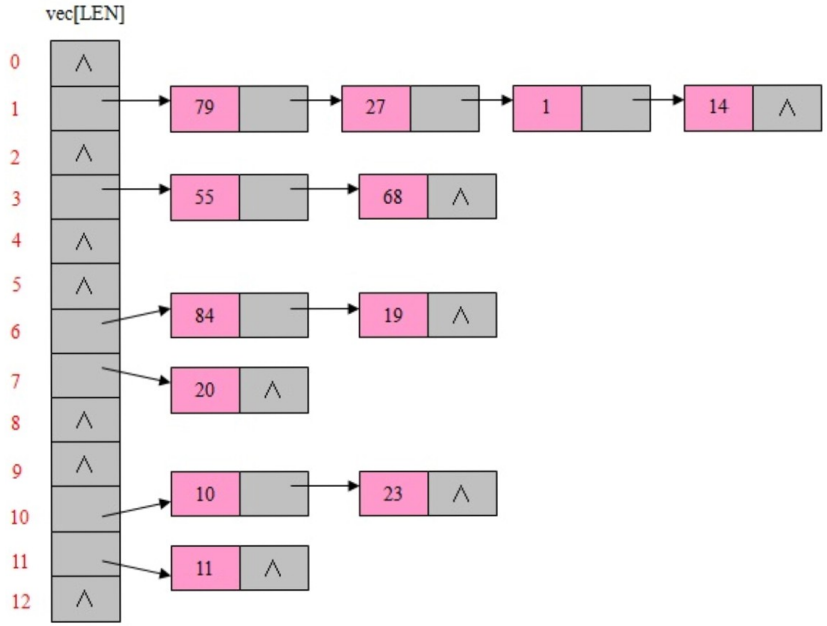
```
//  150.000 string:  vector solution is better with 10%
//       + reserve:                                 15%
// multiset + copy:                                 40%
```

Typical container implementations

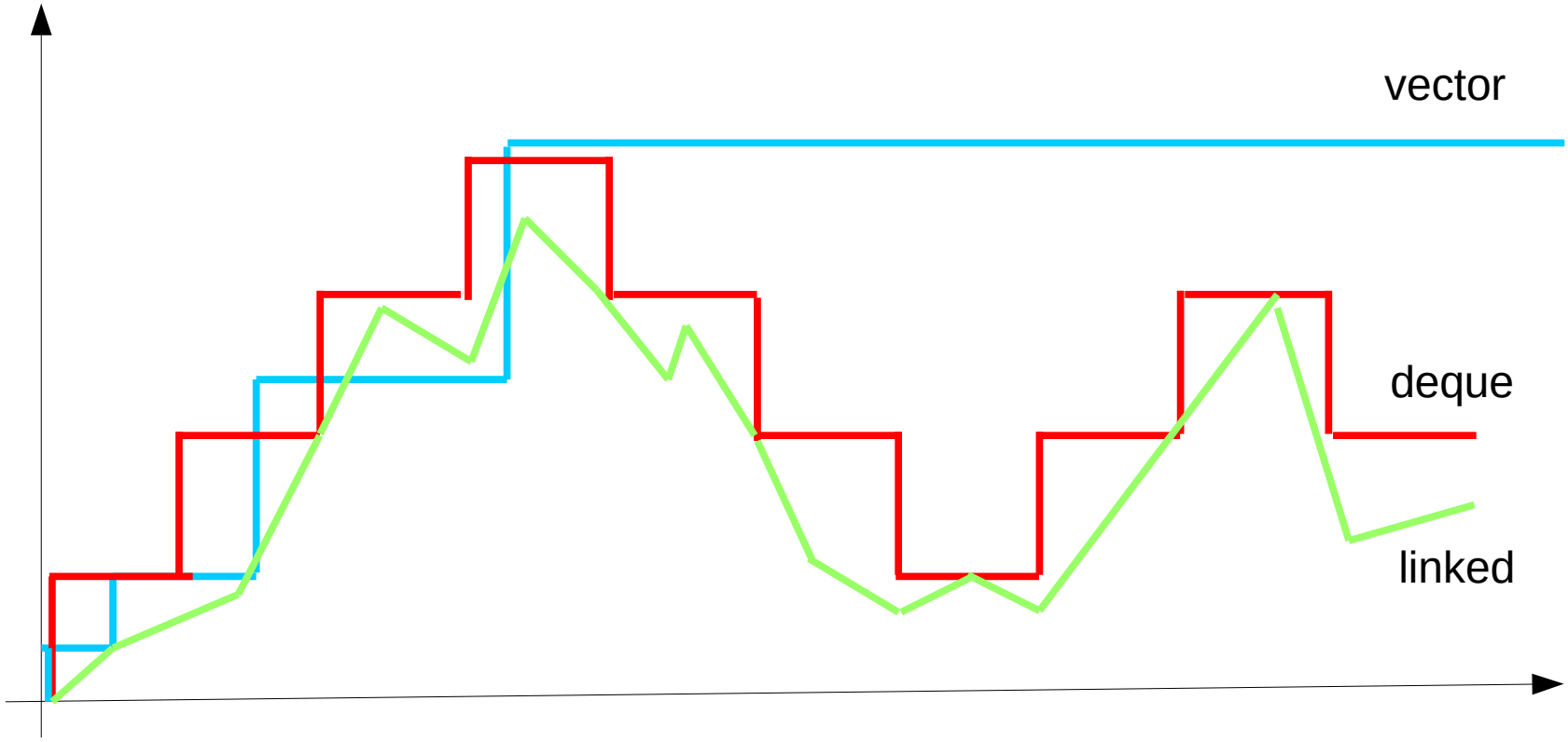








Memory consumption



Vector size and capacity

```
int main()
{
    std::vector<int> v;
    std::cout << "Default-constructed capacity is " << v.capacity() << '\n';

    v.resize(100);
    std::cout << "Capacity of a 100-element vector is " << v.capacity() << '\n';

    v.clear();
    std::cout << "Capacity after clear() is " << v.capacity() << '\n';

    // std::vector<int>(v).swap(v); // C++98

    v.shrink_to_fit(); // C++11
    std::cout << "Capacity after shrink_to_fit() is " << v.capacity() << '\n';
}
```

```
Default-constructed capacity is 0
Capacity of a 100-element vector is 100
Capacity after clear() is 100
Capacity after shrink_to_fit() is 0
```

Iterator invalidation

Category	Container	After **insertion**, are...		After **erasure**, are...		Conditionally
		iterators valid?	**references** valid?	**iterators** valid?	**references** valid?	
Sequence containers	`array`	N/A		N/A		
	`vector`	No		N/A		Insertion changed capacity
		Yes		Yes		Before modified element(s) (for insertion only if capacity didn't change)
		No		No		At or after modified element(s)
	`deque`	No	Yes	Yes, except erased element(s)		Modified first or last element
			No	No		Modified middle only
	`list`	Yes		Yes, except erased element(s)		
	`forward_list`	Yes		Yes, except erased element(s)		
Associative containers	`set` `multiset` `map` `multimap`	Yes		Yes, except erased element(s)		
Unordered associative containers	`unordered_set` `unordered_multiset` `unordered_map` `unordered_multimap`	No	Yes	N/A		Insertion caused rehash
		Yes		Yes, except erased element(s)		No rehash

From: https://en.cppreference.com/w/cpp/container

Std::array

```
#include <string>
#include <iterator>
#include <iostream>
#include <algorithm>
#include <array>

int main()
{
    // construction uses aggregate initialization
    std::array<int, 3> a1{ {1, 2, 3} }; // double-braces required in C++11 (not in C++14)
    std::array<int, 3> a2 = {1, 2, 3};  // never required after =
    std::array<std::string, 2> a3 = { std::string("a"), "b" };

    // container operations are supported
    std::sort(a1.begin(), a1.end());
    std::reverse_copy(a2.begin(), a2.end(), std::ostream_iterator<int>(std::cout, " "));

    std::cout << '\n';

    // ranged for loop is supported
    for(const auto& s: a3)
        std::cout << s << ' ';

    std::array  a1{"foo"};          // C++17 CTAD: std::array<const char*,1>{"foo"}
    auto a2 = std::to_array{"foo"}; // std::array<char,4>{'f','o','o','\n'};
}
```

Std::forward_list

```
template <typename T, typename Alloc = allocator<T> >
class forward_list
{
public:
    void clear();
    iterator insert_after(const_iterator pos, const T& value); // +move, +interval
    iterator emplace_after(const_iterator pos, Args&&... args);
    iterator erase_after(const_iterator pos); // +interval
    void push_front(const T& value);          // +move
    void emplace_front( Args&&... args );
    void pop_front();
    void resise(size_type count); void resise(size_type count, const T& value);
    void swap(forward_list& other);

    iterator before_begin(), begin(), end();
    bool empty();

    void merge(forward_list&& other, Compare comp);
    void splice_after(const_iterator pos, forward_list& other);
    void remove(const T& value);
    void remove_if(UnaryPredicate p);

    void reverse();
    void unique(BinaryPredicate p);
    sort(Compare comp);

     // no reverse iteration
     // no back() or push_back()
     // no size()
};
```

Hash-based containers

- Unordered_map
- Unordered_set
- Unordered_multimap
- Unordered_multiset

std::unordered_map

```
#include <unordered_map>
#include <string>

using namespace std;

int main()
{
    unordered_map<string, string> hashtable;
    hashtable.emplace("www.zolix.hu", "212.92.23.158");
    hashtable.insert(make_pair("www.elte.hu", "212.92.23.159"));

    cout << "IP Address: " << hashtable["www.zolix.com"] << endl;

    for (auto &obj : hashtable)
    {
        cout << obj.first << ": " << obj.second << endl;
    }

    // returns std::unordered_map<std::string,double>::const_iterator
    auto it = hashtable.find("www.elte.com");
    if (hashtable.end() != it)   // hashtable.count("www.elte.com") > 0
    {
        cout << it->first << ": " << it->second << endl;
    }
    return 0;
}
```

std::unordered_map

```
#include <unordered_map>
#include <string>

class MyClass
{
    std::string name;
    int         age;
public:
    Bool operator==(const MyClass& rhs) { ... }   // should be reflexive
    // ...
};
class MyClassHash
{
public:
    size_t operator()(const MyClass& m) const
    {
        return std::hash<std::string>()(m.name) ^ hash<int>()(m.age);
    }
};
int main()
{
    MyClass jim(...), joe(...);
    unordered_map<MyClass, double> salary;
    hashtable.emplace( jim, 20000);
    hashtable.emplace( joe, 22000);
    // ...
}
```

Load factor

- Average insert/find is constant
- Worst-case: linear in container size
- Iterators are invalidated only on rehash
- Load factor == size() / bucket_count() // default 1.0
- Control of buckets:
  - size_type bucket_count() const; // #of buckets
  - float max_load_factor() const; // get max load factor
  - void max_load_factor(float z); // set max load factor
  - size_type bucket_size (size_type n) const; // #of bucket n
  - size_type bucket(const key_t& key) const; // where key goes?
  - void rehash(size_type n); // sets #of buckets

std::unordered_map

```
// unordered_map::bucket_count
#include <iostream>
#include <string>
#include <unordered_map>

int main ()
{
  std::unordered_map<std::string,std::string> mymap = {
       {"house","maison"},{"apple","pomme"},{"tree","arbre"},
       {"book","livre"},{"door","porte"},{"grapefruit","pamplemousse"}
  };
  unsigned n = mymap.bucket_count();
  std::cout << "mymap has " << n << " buckets.\n";

  for (unsigned i=0; i<n; ++i) {
    std::cout << "bucket #" << i << " contains: ";
    for (auto it = mymap.begin(i); it!=mymap.end(i); ++it)
      std::cout << "[" << it->first << ":" << it->second << "] ";
    std::cout << "\n";
  }
}
mymap has 7 buckets.
bucket #0 contains: [book:livre] [house:maison]
bucket #1 contains:
bucket #2 contains:
bucket #3 contains: [grapefruit:pamplemousse] [tree:arbre]
bucket #4 contains:
bucket #5 contains: [apple:pomme]
bucket #6 contains: [door:porte]
```

Load factor

```
// unordered_map::max_load_factor
int main ()
{
  std::unordered_map<std::string,std::string> mymap = {
    {"Au","gold"},{"Ag","Silver"},{"Cu","Copper"},{"Pt","Platinum"}
  };

  std::cout << "current max_load_factor: " << mymap.max_load_factor() << std::endl;
  std::cout << "current size: " << mymap.size() << std::endl;
  std::cout << "current bucket_count: " << mymap.bucket_count() << std::endl;
  std::cout << "current load_factor: " << mymap.load_factor() << std::endl;

  float z = mymap.max_load_factor();
  mymap.max_load_factor ( z / 2.0 );

  std::cout << "new max_load_factor: " << mymap.max_load_factor() << std::endl;
  std::cout << "new size: " << mymap.size() << std::endl;
  std::cout << "new bucket_count: " << mymap.bucket_count() << std::endl;
  std::cout << "new load_factor: " << mymap.load_factor() << std::endl;
  return 0;
}
current max_load_factor: 1
current size: 4
current bucket_count: 5
current load_factor: 0.8
new max_load_factor: 0.5
new size: 4
new bucket_count: 11
new load_factor: 0.363636
```

Std::unordered_multi...

```
// unordered_multiset::equal_range

#include <iostream>
#include <string>
#include <unordered_set>

int main ()
{
  std::unordered_multiset<std::string> myums =
                        {"cow","pig","pig","chicken","pig","chicken"};

  auto myrange = myums.equal_range("pig");

  std::cout << "These pigs were found:";

  while ( myrange.first != myrange.second ) {
      std::cout << " " << *myrange.first++;
  }
  std::cout << std::endl;

  return 0;
}
```

Example: word counter

```
#include <cctype>
#include <iterator>
#include <iostream>
#include <vector>
#include <algorithm>



int main()
{
  std::vector<char> v{};
  std::cin >> std::noskipws;
  std::copy( std::istream_iterator<char>{std::cin}, std::istream_iterator<char>{},
                                                                  std::back_inserter(v));
  int  cnt = 0;
  char prev = '\n';  // the imaginary char on -1 position is a white space.

  for ( char curr : v )
  {
    if ( std::isspace(prev) && !std::isspace(curr) )  // new word starts
      ++cnt;
    prev = curr;
  }
  std::cout << cnt << '\n';
}
```

Example: word counter

```
#include <cctype>
#include <iterator>
#include <iostream>
#include <vector>
#include <algorithm>
#include <numeric>


int main()
{
  std::vector<char> v{};
  std::cin >> std::noskipws;
  std::copy( std::istream_iterator<char>{std::cin}, std::istream_iterator<char>{},
                                                                  std::back_inserter(v));
  int cnt = 0;
  if ( ! v.empty() )
  {
    cnt = std::transform_reduce(
       std::begin(v), std::end(v)-1, std::begin(v)+1,!std::isspace(v[0])?1:0, std::plus{},
       [](char curr, char next){ return std::isspace(curr) && !std::isspace(next); } );

    std::cout << cnt << '\n';
}
```

Example: word counter

```
#include <cctype>
#include <iterator>
#include <iostream>
#include <vector>
#include <algorithm>
#include <numeric>
#include <execution>

int main()
{
  std::vector<char> v{};
  std::cin >> std::noskipws;
  std::copy( std::istream_iterator<char>{std::cin}, std::istream_iterator<char>{},
                                                                std::back_inserter(v));
  int cnt = 0;
  if ( ! v.empty() )
  {
    cnt = std::transform_reduce( std::execution::par,
       std::begin(v), std::end(v)-1, std::begin(v)+1,!std::isspace(v[0])?1:0, std::plus{},
       [](char curr, char next){ return std::isspace(curr) && !std::isspace(next); } );

    std::cout << cnt << '\n';
}
```

par_algorithms (C++17)

- Based on Intel's Threading Building Blocks (TBB)
- Extends STL algorithms with execution policy
  - std::execution::seq Sequential execution
  - std::execution::par Parallel execution
  - std::execution::par_unseq Parallel SIMD execution
  - std::execution::unseq Sequential SIMD execution
- These policies are permissions not obligations. Implementation may choose what can be parallelized
- Minimal requirement: forward iterator
- The programmer's task to ensure that element access functions will not cause dead lock or data race
- In case of paralellization and vectorization access must not use any blocking synchronization

Parallel STL

```
// Example from Stroustrup

template<class T, class V>
struct Accum     // simple accumulator function object
{
    T* b;
    T* e;
    V val;
    Accum(T* bb, T* ee, const V& vv) : b{bb}, e{ee}, val{vv} {}
    V operator() () { return std::accumulate(b,e,val); }
};

double comp(vector<double>& v)   // spawn many tasks if v is large enough
{
    if (v.size()<10000) return std::accumulate(v.begin(),v.end(),0.0);

    auto f0 {async(Accum{&v[0],&v[v.size()/4],0.0})};
    auto f1 {async(Accum{&v[v.size()/4],&v[v.size()/2],0.0})};
    auto f2 {async(Accum{&v[v.size()/2],&v[v.size()*3/4],0.0})};
    auto f3 {async(Accum{&v[v.size()*3/4],&v[v.size()],0.0})};

    return f0.get()+f1.get()+f2.get()+f3.get();
}
```

Parallel STL

```
// Example from cppreference

template<class T, class V>
struct Accum     // simple accumulator function object
{
    T* b;
    T* e;
    V val;
    Accum(T* bb, T* ee, const V& vv) : b{bb}, e{ee}, val{vv} {}
    V operator() () { return std::accumulate(b,e,val); }
};

double comp(vector<double>& v)
{
    // non-deterministic if binary_op is not associative or not commutative
    double res = std::reduce(std::execution::par, v.begin(), v.end(), 0.0);
    return res;
}
```

Vectorization

```
std::vector<int> v {1,2, ... };

int sum { std::accumulate(v.begin(), v.end(), 0) };


int sum = 0;
for ( size_t i = 0; i < v.size(); ++i)
{
    sum += v[i];
}



int sum = 0;
for ( size_t i = 0; i < v.size() / 4; i+=4)
{
    sum += v[i] + v[i+1] + v[i+2] + v[i+3];   // most CPU supports this
}
// handle if (v.size()/4) is not 0
```

accumulate() vs reduce()

```
#include <iostream>
#include <vector>

int main()
{
  std::vector<long long> v1;
  for ( int i = 0; i < 10; ++i)
  {
    v1.insert( v1.end(), {0,1,2,3,4}); // creates 50 elements
  }

  long long sum = 0;
  for ( std::size_t i = 0; i < v1.size(); ++i) // summa x^2  x in [0..49]
  {
    sum += v1[i]*v1[i];
  }

  std::cout << sum << '\n';

  return 0;
}
```

```
$ ./a.out
300
```

accumulate() vs reduce()

```
#include <iostream>
#include <numeric>
#include <vector>


std::vector<long long> v1;
auto sqrsum = [] (auto s, auto val) { return s + val * val; };

int main()
{
  for ( int i = 0; i < 10; ++i)
  {
    v1.insert( v1.end(), {0,1,2,3,4});  // creates 50 elements
  }

  auto sum1 = std::accumulate(v1.begin(), v1.end(), 0LL, sqrsum);  // classical STL



  std::cout << sum1 << '\n';
  return 0;
}


$ ./a.out
300
```

accumulate() vs reduce()

```
#include <iostream>
#include <numeric>
#include <vector>


std::vector<long long> v1;
auto sqrsum = [] (auto s, auto val) { return s + val * val; };

int main()
{
  for ( int i = 0; i < 10; ++i)
  {
    v1.insert( v1.end(), {0,1,2,3,4});  // creates 50 elements
  }
  // accumulate is guaranteed left associative
  auto sum1 = std::accumulate(v1.begin(), v1.end(), 0LL, sqrsum);  // classical STL



  std::cout << sum1 << '\n';
  return 0;
}


$ ./a.out
300
```

accumulate() vs reduce()

```
#include <iostream>
#include <numeric>
#include <vector>
#include <execution>

std::vector<long long> v1;
auto sqrsum = [] (auto s, auto val) { return s + val * val; };

int main()
{
  for ( int i = 0; i < 10; ++i)
  {
    v1.insert( v1.end(), {0,1,2,3,4});
  }
  // accumulate is guaranteed left associative
  auto sum1 = std::accumulate(v1.begin(), v1.end(), 0LL, sqrsum);
  // reduce can work parallel
  auto sum2 = std::reduce(std::execution::par, v1.begin(), v1.end(), 0LL, sqrsum);

  std::cout << sum1 << “, “ << sum2 << '\n';
  return 0;
}
```

```
$ ./a.out
300, 300
```

accumulate() vs reduce()

```
#include <iostream>
#include <numeric>
#include <vector>
#include <execution>

std::vector<long long> v1;
auto sqrsum = [] (auto s, auto val) { return s + val * val; };

int main()
{
  for ( int i = 0; i < 1000; ++i)
  {
    v1.insert( v1.end(), {0,1,2,3,4});
  }
  // accumulate is guaranteed left associative
  auto sum1 = std::accumulate(v1.begin(), v1.end(), 0LL, sqrsum);
  // reduce can work parallel
  auto sum2 = std::reduce(std::execution::par, v1.begin(), v1.end(), 0LL, sqrsum);

  std::cout << sum1 << “, “ << sum2 << '\n';
  return 0;
}
```

```
$ ./a.out
30000, 30000
```

accumulate() vs reduce()

```
#include <iostream>
#include <numeric>
#include <vector>
#include <execution>

std::vector<long long> v1;
auto sqrsum = [] (auto s, auto val) { return s + val * val; };

int main()
{
  for ( int i = 0; i < 1000000; ++i)
  {
    v1.insert( v1.end(), {0,1,2,3,4});
  }
  // accumulate is guaranteed left associative
  auto sum1 = std::accumulate(v1.begin(), v1.end(), 0LL, sqrsum);
  // reduce can work parallel
  auto sum2 = std::reduce(std::execution::par, v1.begin(), v1.end(), 0LL, sqrsum);

  std::cout << sum1 << “, “ << sum2 << '\n';
  return 0;
}
```

```
$ ./a.out
30000000, 59820950156796
```

accumulate() vs reduce()

```
#include <iostream>
#include <numeric>
#include <vector>
#include <execution>

std::vector<long long> v1;
auto sqrsum = [] (auto s, auto val) { return s + val * val; };  // not commutative

int main()
{
  for ( int i = 0; i < 1000000; ++i)
  {
    v1.insert( v1.end(), {0,1,2,3,4});
  }
  // accumulate is guaranteed left associative
  auto sum1 = std::accumulate(v1.begin(), v1.end(), 0LL, sqrsum);
  // reduce can work parallel
  auto sum2 = std::reduce(std::execution::par, v1.begin(), v1.end(), 0LL, sqrsum);

  std::cout << sum1 << “, “ << sum2 << '\n';
  return 0;
}
```

```
$ ./a.out
30000000, 59820950156796
```

accumulate() vs reduce()

```
#include <iostream>
#include <numeric>
#include <vector>
#include <execution>
#include <functional>
std::vector<long long> v1;
auto sqrsum = [] (auto s, auto val) { return s + val * val; };

int main()
{
  for ( int i = 0; i < 1000000; ++i)
  {
    v1.insert( v1.end(), {0,1,2,3,4});
  }
  // accumulate is guaranteed left associative
  auto sum1 = std::accumulate(v1.begin(), v1.end(), 0LL, sqrsum);
  auto sum2 = std::transform_reduce(std::execution::par,  // map-reduce
          v1.begin(), v1.end(), 0LL,
          std::plus<>(),
          [](auto v) { return v*v; });
  std::cout << sum1 << “, “ << sum2 << '\n';
  return 0;
}

$ ./a.out
30000000, 30000000
```